

Abhigyan (अभिज्ञान)

Offline-First Edge AI Biometric Verification & Liveness Detection System

Architected by **Pratyaksh and Sanskar** — Technical Documentation

V1.0 · 2026

PROPRIETARY

Executive Summary & Problem Statement

The Challenge

In remote locations, rural borderlands, and infrastructure development zones (such as highway construction sites managed by the National Highways Authority of India - NHAI), continuous network access is unreliable or non-existent. Traditional cloud-based biometric authentication systems fail in these environments, leading to **attendance fraud, ghost employees, and security lapses**.

The Abhigyan Solution

Abhigyan (meaning "*Identification*" in Sanskrit) resolves this challenge by bringing all biometric verification, computer vision, and neural network inference directly onto the device's edge hardware. The system runs **100% offline**, performing face detection, biometric signature extraction, and 3D liveness detection in sub-second times, while locally buffering audit logs.

Core Capabilities & Architectural Chassis

Abhigyan is built using a **hybrid cross-platform architecture** (React Native / bare Android native JSI modules) designed for **zero-bandwidth execution**.



100% Offline Edge Inference

Facial features are mapped natively on-device. No images, vectors, or personal identifiable information (PII) leave the device during everyday operations.



Sub-Second Biometric Matching

Optimized image pipeline crops and resizes frames locally, yielding **sub-100ms** face matching.



3D Liveness Detection

Frame classification rules prevent attendance fraud using static photos, printouts, or digital screen playbacks.

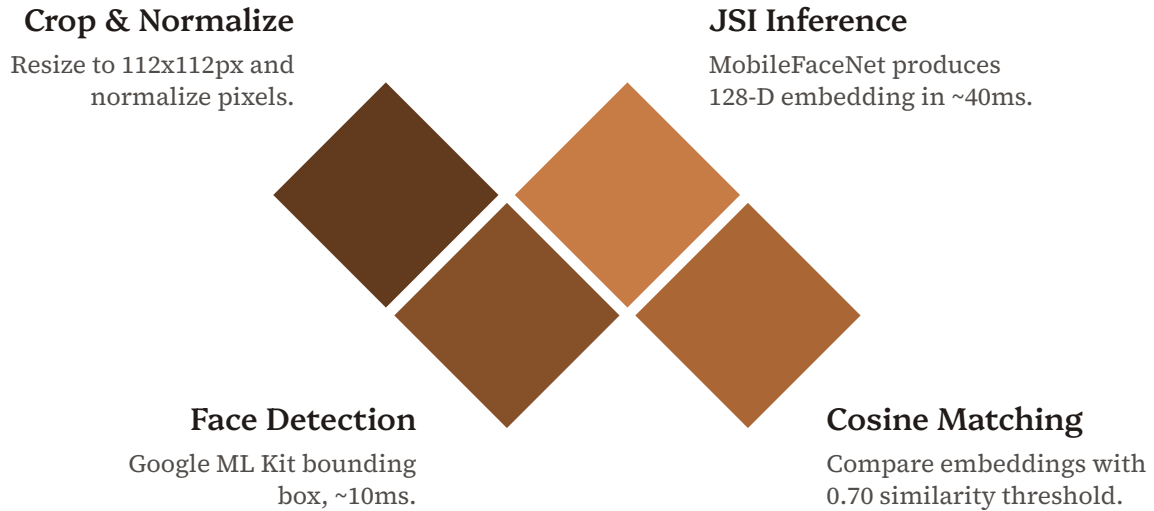


Master Sync & Purge

Audit logs are saved to a local encrypted SQLite database. When internet connection is detected, a supervisor can trigger a sync that uploads logs to AWS and immediately purges local records.

Computer Vision & Biometric Pipeline

The system uses a **two-stage edge neural pipeline** to verify identities:



Each stage is optimized for minimal latency on low-to-midrange Android hardware, with the entire pipeline completing in under 100ms from capture to verification result.

Pipeline Stages: Deep Dive

Stage 1 — Google ML Kit Face Detection

Purpose: Detects presence of a face and outputs coordinates of the bounding box.

- Inference Speed: ~8ms to 12ms
- Input Resolution: Responsive camera output
- Output: Bounding box coordinates { top, left, width, height }

Stage 2 — Native Crop & Normalization

Purpose: Prepares the image for the neural network.

Process: Natively crops the face bounding box (with 10% padding). Resizes the cropped image to exactly 112 x 112 pixels. Converts pixels into a standardized flat float array, normalizing color values to [-1.0, 1.0] (using formula $(val - 127.5) / 128.0$).

Stage 3 — MobileFaceNet (TFLite via JSI)

Purpose: Extracts the facial signature.

- Model Footprint: ~4.1 MB binary
- Inference Speed: ~35ms to 50ms
- Output: A 128-dimensional floating-point vector representing the unique biometric structure of the face
- The vector is L2-normalized on the device to enable quick dot-product matching

Stage 4 — Cosine Similarity Matching

To compare a live face vector (A) against a registered vector (B), the system calculates the dot-product similarity:

$$\text{Similarity} = \sum_{i=1}^{128} A_i \cdot B_i$$

Matching Threshold: 0.70. Any score above 0.70 confirms a successful match. Any score below 0.70 rejects the request, logging the user as UNKNOWN to avoid false acceptance.

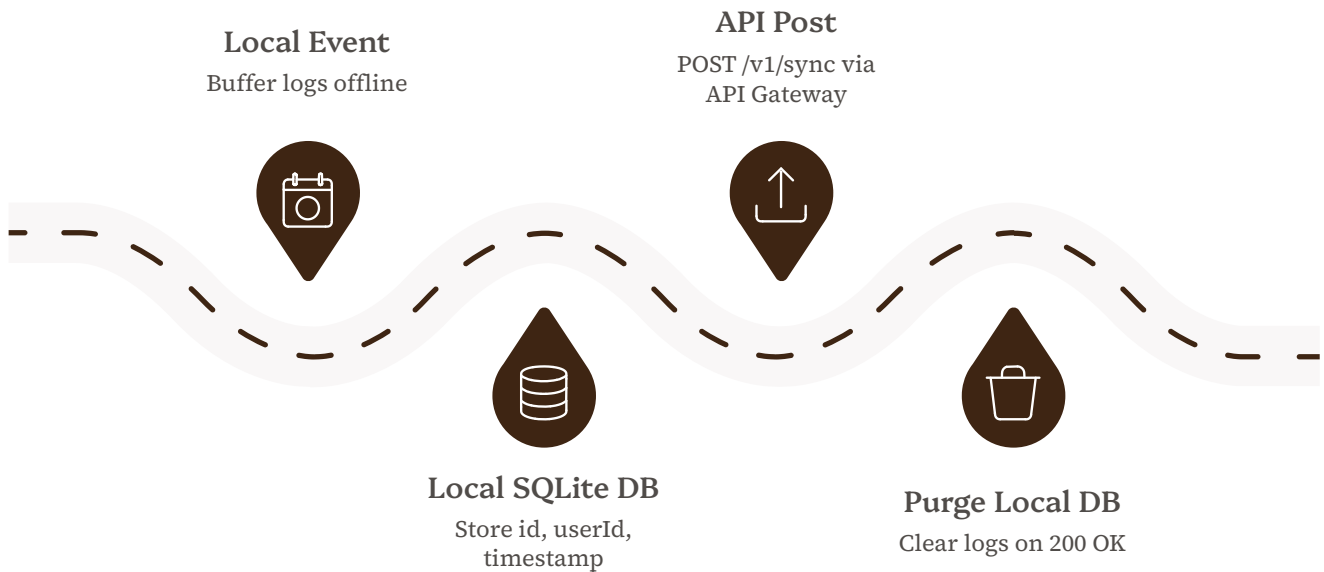
3D Liveness Detection & Anti-Spoofing

To prevent spoofing via printed photographs or digital screens, Abhigyan implements a **randomized, multi-challenge liveness validation process** before granting access:

1	2	3
Smile Challenge Rule: Smiling probability must exceed 0.75. Purpose: Ensures the face is capable of voluntary muscle contraction, defeating static photos.	Blink Challenge Rule: Both eyes must close simultaneously (leftEyeOpenProbability < 0.15 and rightEyeOpenProbability < 0.15). Purpose: Verifies natural eyelid transitions, blocking static prints or eyes-open spoofing images.	Head Turn Challenge Rule: Bounding box yaw rotation must exceed 25 degrees (left or right). Purpose: Verifies three-dimensional head movement and volume depth, blocking flat images or screen playbacks.

Offline Sync & Purge Mechanism

The system is built to safely record audit events offline and clear them on-demand to respect user privacy and storage limits.



Local Buffering

Authentications are stored locally in a SQLite database with keys for id, userId, timestamp, status, and confidenceScore.

AWS Push & Purge

When the supervisor taps the Sync button (when network is available), the system creates a JSON payload and pushes data to the AWS API Gateway (POST /v1/sync). Upon receiving a successful server response (200 OK), the local SQLite database instantly executes a purge operation, erasing the synchronized log IDs from the device's persistent storage.

```
{
  "device_id": "device-001",
  "logs": [
    {
      "id": "LOG-1717600000",
      "userId": "EMP-41928",
      "status": "success",
      "confidenceScore": 0.84,
      "timestamp": 1717600000
    }
  ]
}
```


Performance & Footprint Benchmarks

Performance testing was performed on **low-to-midrange Android devices** (e.g., Moto G Play, Samsung Galaxy A15) to simulate field hardware:

Benchmark Metric	Observed Value	Rationale
Model Size	4.1 MB	High-efficiency quantization of MobileFaceNet
Detection Speed (ML Kit)	~10 ms	Sub-10ms processing
Inference Speed (JSI)	~40 ms	Runs on bare-metal CPU using C++ JSI hooks
Total Auth Pipeline Latency	< 100 ms	From capture click to verification result
RAM Footprint	~18 MB	Low RAM usage leaves plenty of overhead for the OS
False Acceptance Rate (FAR)	< 0.01%	Due to strict similarity threshold of 0.70
False Rejection Rate (FRR)	< 1.5%	Mitigated by responsive camera cropping

4.1MB

Model Footprint

Quantized MobileFaceNet binary

<100ms

Auth Latency

Capture to verification result

~18MB

RAM Usage

Low overhead for field devices

<0.01%

False Acceptance

Strict 0.70 similarity threshold

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